Experiment : 5

Date:

# USE CASE DIAGRAM

**Aim:**

To understand and apply software engineering concepts by designing a class diagram and a time sequence diagram. The objective of this experiment is to represent both the structural and behavioural aspects of a system using standard UML notations. Through this activity, students learn how system components interact with each other and how data flows between different modules during execution**.**

**Introduction**

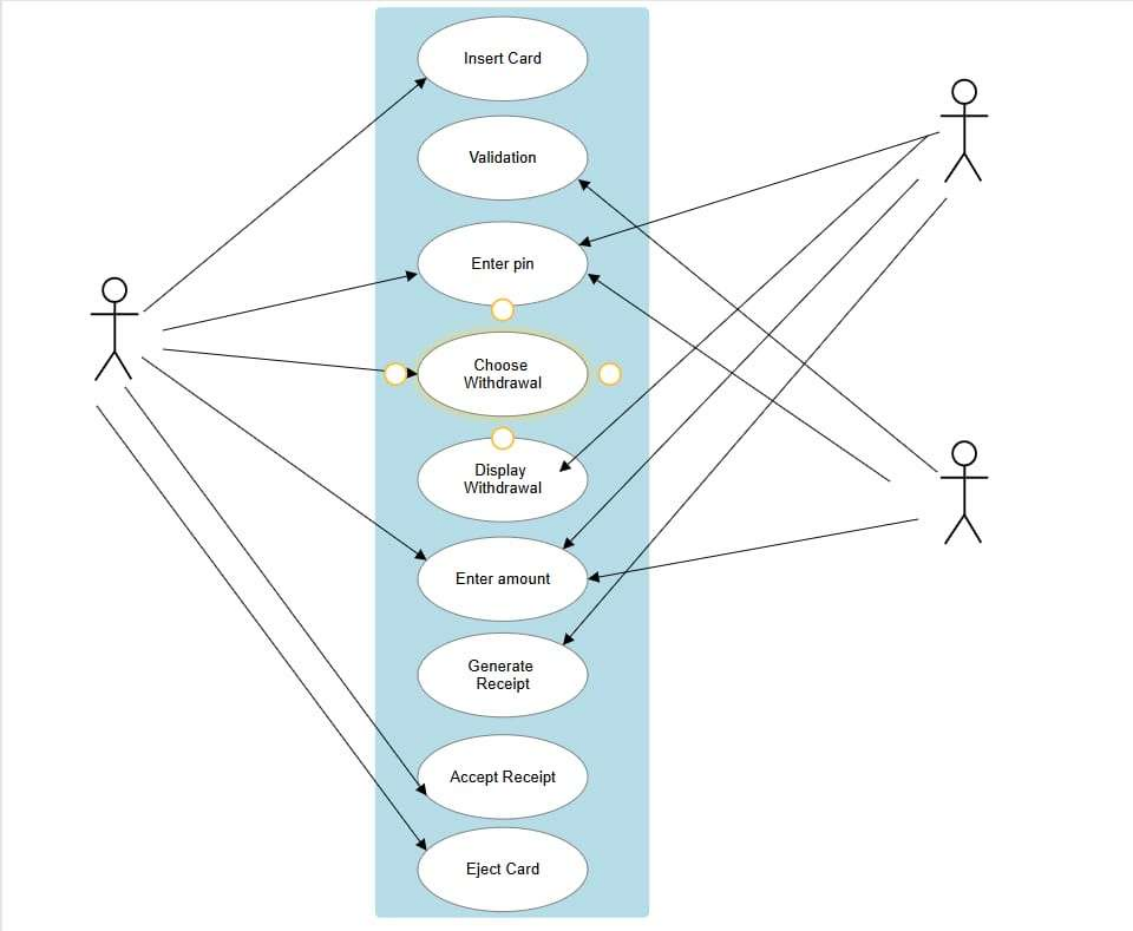
Automated Teller Machines (ATMs) are electronic banking systems that allow customers to perform financial transactions without visiting a bank branch. A use-case diagram is a UML behavioral diagram that represents system functionality from the user's perspective. It shows how different actors interact with the system to complete tasks.

In the ATM withdrawal system, the customer communicates with the ATM interface, which in turn connects to the bank server for authentication and transaction processing. The diagram illustrates major actions such as card insertion, PIN validation, withdrawal selection, amount entry, receipt generation, and card ejection.

**Actors Involved**

1. Customer/User – Initiates and performs withdrawal actions.
2. ATM Machine – Provides the interface for user interaction.
3. Bank Server/System – Validates credentials and processes transactions securely.

USE CASE DIAGRAM:




**Pre-Conditions**

- The ATM machine must be powered on and connected to the banking network.
- The user must possess a valid ATM card.
- The user must know the correct PIN.
- The account should have sufficient balance for withdrawal.

**Observation**

1. The user inserts the ATM card into the machine.
2. The ATM reads the card information and sends it for validation.
3. The system prompts the user to enter the PIN.

4. The entered PIN is verified with the bank server database.
5. After successful authentication, the system displays transaction options.
6. The user selects the Withdrawal option from the menu.
7. The ATM displays available withdrawal choices.
8. The user enters the desired withdrawal amount.
9. The system checks account balance and transaction limits.
10. If sufficient funds are available, the ATM processes the withdrawal request.
11. The ATM dispenses the cash to the user.
12. The system generates a transaction receipt.
13. The user chooses whether to accept the receipt.
14. Finally, the ATM ejects the card and ends the session.

**Exception Handling**

- If the PIN entered is incorrect, the system asks for re-entry and may block the card after multiple failed attempts.
- If the withdrawal amount exceeds the available balance, the system displays an error message.
- If the ATM runs out of cash or network fails, the transaction is cancelled automatically.
- If the user cancels the transaction midway, the ATM ejects the card safely.

**Post-Conditions**

- Transaction details are stored in the banking database.
- Account balance is updated after successful withdrawal.
- User receives cash and optional receipt.
- ATM session is terminated securely.

**Advantages Observed**

- Provides secure authentication through PIN validation.
- Enables quick and convenient cash withdrawal.
- Reduces dependency on manual banking services.
- Maintains accurate transaction records.

**Conclusion**

The ATM Withdrawal Use-Case Diagram clearly represents the interaction between the customer, ATM machine, and bank server. The system follows a structured flow starting from card insertion to card ejection while ensuring security, validation, and proper transaction handling. This observation helps in understanding real-world banking system design using UML concepts.